Lab 7: Fuzz Testing for Input Validation Bugs

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```
==== Running first_character_type ====
Enter a password: alice123
First Character Type: letter

==== Running split_credentials ====
Enter user:password (e.g. alice:Secret1):  alice:Secret1
User and Password: ('alice', 'Secret1')

==== Running digit_ratio ====
Enter a password to measure: Pass123
Digit Ratio: 0.42857142857142855
$ python3 password_policy.py
==== Running first_character_type ====
Enter a password:
Traceback (most recent call last):
  File "/Users/sam/sem-5/se/Lab/Lab-7/password_policy.py", line 33, in <module>
    main()
  File "/Users/sam/sem-5/se/Lab/Lab-7/password_policy.py", line 19, in main
    result = first_character_type(user_input)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/Users/sam/sem-5/se/Lab/Lab-7/password_policy.py", line 2, in first_charac
ter_type
    ch = pw[0]
         ~~^^^
IndexError: string index out of range
$
```

Figure 1. Step 2: output of the buggy program with a normal password and then an empty one.

```
test_digit_ratio_no_crash
    |     digit_ratio(s)
    |   File "/Users/sam/sem-5/se/Lab/Lab-7/password_policy.py", line 14, in digit
_ratio
    |     return sum(c.isdigit() for c in pw) / len(pw)
    |                ^^^^^^^^^^^^^^^^^^^^^^^^^
    | TypeError: 'NoneType' object is not iterable
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/sam/sem-5/se/Lab/Lab-7/test_password_policy.py", line 23, in
test_digit_ratio_no_crash
    |     assert False, f"digit_ratio crashed with: {e}"
    | AssertionError: digit_ratio crashed with: 'NoneType' object is not iterable
    | assert False
    | Failing test case: test_digit_ratio_no_crash(
    |     s=None,
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         /Users/sam/sem-5/se/Lab/Lab-7/test_password_policy.py:22
    +------------------------------------
=========================== short test summary info ============================
FAILED test_password_policy.py::test_first_character_type_no_crash - ExceptionG...
FAILED test_password_policy.py::test_split_credentials_no_crash - ExceptionGrou...
FAILED test_password_policy.py::test_digit_ratio_no_crash - ExceptionGroup: Hyp...
============================== 3 failed in 0.44s ===============================
$
```

Figure 2. Step 4: failing Hypothesis tests on the buggy code.

```
$ python3 password_policy.py
==== Running first_character_type ====
Enter a password:
First Character Type: empty

==== Running split_credentials ====
Enter user:password (e.g. alice:Secret1):
User and Password: None

==== Running digit_ratio ====
Enter a password to measure:
Digit Ratio: 0.0
$
```

Figure 3. Step 5: output of the fixed program with empty inputs.

```
$ pytest test_password_policy.py
============================== test session starts ==============================
platform darwin -- Python 3.12.1, pytest-9.1.1, pluggy-1.5.0
rootdir: /Users/sam/sem-5/se/Lab/Lab-7
plugins: hypothesis-6.170.14, platformdirs-4.13.5, cov-7.1.0, anyio-3.7.1, xonsh-0
.18.3
collected 3 items

test_password_policy.py ...                                              [100%]

============================== 3 passed in 0.38s ===============================
$
```

Figure 4. Step 6: passing Hypothesis tests on the fixed code.